Sahaj Bhadja

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Summary

Software engineer with 2+ years building **traceable engineering workflows**, **risk-aware validation systems**, and **automation-first release controls** for safety-critical, high-change software environments.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 - Present*

- Built **structured workflow orchestration** and service interfaces for 500+ users, reducing latency from **1.2s to 350ms** while improving traceability of requirements-to-execution paths.
- Implemented **validation gates and rollback controls** for 5K+ monthly jobs, increasing release confidence and enabling audit-ready evidence for change-impact review.

Grad8 **LA, California**
Software Engineer Intern *June 2025 - August 2025*

- Developed **full-stack production modules** handling 10K+ daily requests, converting software requirements into testable API/UI components with controlled deployment behaviors.
- Improved **CI/CD compliance checkpoints** and automated defect triage, reducing release turnaround by **60%** while preserving quality under frequent design updates.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 - May 2025*

- Built **verification pipelines** across 100K+ trajectory logs, enabling reproducible evaluation baselines and measurable risk-aware analysis for safety-sensitive experimentation.
- Generated **technical diagnostics and evidence summaries** to support review readiness, helping teams map findings to implementation plans and release decisions.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 - July 2024*

- Engineered **API/ETL services** with Python, AWS Glue, and S3, improving response time by **30%** while strengthening controlled data flow across distributed systems.
- Maintained **20+ automated test suites** using Jenkins, PyTest, and Mockito, improving pipeline efficiency by **68%** and reducing recurring defects in production releases.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 - August 2023*

- Implemented **MERN + WebSocket workflows** for eight modules and 2M+ synchronized records, improving consistency and operational traceability in high-concurrency scenarios.
- Automated **containerized deployment and validation** pipelines via Docker/Jenkins, reducing rollout time by **50%** with repeatable release and recovery processes.

Skills

Languages: **Python**, **SQL**, C/C++, Java, JavaScript/TypeScript, Bash
Design Controls: requirements traceability, verification/validation mapping, change control, release evidence generation
Risk & Quality: risk-control linkage, defect management, audit-readiness mindset, safety-critical documentation
Engineering Systems: CI/CD pipelines, automated test frameworks, controlled deployment workflows, Linux/UNIX environments
Tooling: Git/GitHub, Jenkins, Docker, Agile/Scrum collaboration, cross-functional communication

Projects

Software Traceability Matrix Engine | *Python, SQL, FastAPI, Docker*

- Built an **end-to-end traceability service** linking user needs, requirements, risk controls, verification cases, and release artifacts, improving review speed and compliance visibility.

Chat2SQL Code | *Java, TypeScript, SQL, Docker*

- Developed a **schema-aware query assistant** across 10+ datasets with rule-based validation and explainable outputs, strengthening requirement-to-query correctness and troubleshooting speed.

Linguistix Code | *Python, AWS, Docker, CI/CD*

- Implemented a **containerized REST data pipeline** with automated CI/CD validation, improving reliability, controlled releases, and operational continuity for production workloads.